VF Asia Footwear Laboratory Testing Manual

Test Method for Compression Set (Static)	Test method # VF- B10 REF #: TM-R- 009 Page 1 of 2
Date Issued: May 23, 2017 Revision #: 1 Reason for revision: 1 Only the soft PU used as backers, tongue and counter padding should be tested at 23 ± 2 ºC for 22 hours. 2. Change to digital caliper from thickness gauge for checking the thickness.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the ability of foam to return to its original thickness after being compressed/deflected between two parallel plates at a specified temperature and duration.

REFERENCES: ASTM D3574

EQUIPMENT AND MATERIAL:

1. Compression platen, meeting the requirements of ASTM D 3574
2. Spacers of thickness 4.0, 4.5, 5.0, 5.5, 6.0 mm.
3. Digital calipers capable of measuring up to 0.01mm
4. Air-circulating oven, capable of maintaining 45.0 ± 2 ºC
5. Expanded Polymer or Foam

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory conditions @ 23.0 ± 2.0 ˚C temperature and 55 ±10 % Relative Humidity.

2. Die cut four test samples having a diameter of 2.54 cm and a thickness of 10.0 ± 2.0 mm. The sample must be prepared with parallel top and bottom surfaces, and sides which are perpendicular to the top and bottom surfaces. If a test sample is less than 8.0 mm, multiple layers of the test material must be stacked, **without the use of adhesive**, until a thickness of 10.0 ± 2.0 mm is obtained.

REPETITIONS: Test four (4) samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0± 2.0 ˚C temperature and 55 ± 10 % Relative Humidity.

1. Draw 2 lines – could be parallel or across to each other.
2. Mark each line – ensure that the mark of each line is unique from the other line.
3. Measure the thickness of the sample, thus each specimen has 2 thickness readings and 2 results of compression set.
4. Place the appropriate spacers on the compression platen, such that 50 % compression will be obtained, as follows:

Sample thickness	Spacer thickness
8.0-8.5 mm	4.0 mm
8.5-9.5 mm	4.5 mm

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9.5-10.5 mm	5.0 mm
10.5-11.5 mm	5.5 mm
11.5-12.0 mm	6.0 mm

3. Place the test specimen onto the bottom platen and position the top platen onto the test specimen. Tighten the nuts on the test equipment until the top platen firmly contacts each of the spaces.

4. Within 15minutes after compressing the samples, place the compression set equipment with test samples into their appropriate environment as described below:

- EVA, Latex, Polyethylene, Polyurethane = 45 ± 2º C for 6 hours
- Polyurethane – soft type utilized as backers, for Tongue and counter padding = 23 ± 2 ºC for 22 hours

5. At completion of the exposure, immediately remove the test sample from the compression set equipment.

6. Allow the materials to recover at 23+/-2 ºC as described below:

- EVA, Latex, Polyethylene, Polyurethane = 30 ± 2 minutes
- Polyurethane - soft type utilized as backer for Tongue and counter padding = 60 ± 2 minutes

7. Immediately following the completion of the recovery period, measure the thickness of the samples, in the same manner described in **STEP 3**. If the sample has set unevenly, record the minimum thickness.

CALCULATION:

% Compression set = $\frac{\text{Original thickness – Final thickness}}{\text{Original thickness - Spacer thickness}}$ * 100

REPORT: The average of the four test samples to the nearest 0.1%.